

RAVI RANJAN PRASAD

rrk794063@gmail.com · +91 9142162772 · GitHub · LinkedIn

EDUCATION

B.Tech, Computer Science & Engineering

Haridwar University | Jun 2027

B.S, Data Science

IIT Madras | Jun 2027

Guru Gobind Singh Public School

Class X · Class XII | 2020 / 2022

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript

Backend & APIs: Spring Boot, REST APIs, Spring Security, JWT, BCrypt, Spring Data JPA, gRPC (Protobuf/HTTP/2)

ML: PyTorch, YOLOv8, OpenCV, Scikit-learn, MediaPipe, Hugging Face Transformers, NumPy, Pandas, Matplotlib, Seaborn

Databases: MySQL, PostgreSQL, SQL

CS Fundamentals: DSA, OOP, DBMS, Computer Networks, Distributed Systems, Consistent Hashing, Raft Consensus

Core Java: Collections Framework, Generics, Stream API, Concurrency

Testing: JUnit, Mockito, MockMvc

Tools: Git, Maven, VS Code, Jupyter Notebook, Google Colab, Ollama, Arduino (PyFirmata2), Tkinter

PROJECTS

HealFS — Self-Healing Distributed File System

Java · gRPC (Protobuf/HTTP/2) · Raft Consensus · Consistent Hashing · JUnit/Mockito

- Architected a multi-node distributed file system in pure Java with replication factor 3 via gRPC (Protobuf/HTTP/2), achieving production-grade fault tolerance with no single point of failure.
- Implemented consistent hashing with 150 virtual nodes per physical node, balancing chunk distribution across DataNodes and eliminating hotspots at scale.
- Built a heartbeat-based ALIVE→SUSPECTED→DEAD failure detector with automatic re-replication, restoring the 3× replication factor within seconds of any node failure.
- Designed Raft consensus with term-based voting and randomised election timeouts, guaranteeing single-master leader election with zero split-brain scenarios.
- Enforced SHA-256 checksum validation on every chunk write and replication; throttled concurrent re-replications via Semaphore to prevent cluster flooding.
- Achieved high test coverage across consistent hashing, replication manager, and Raft election logic via JUnit/Mockito unit and integration tests.

DisasterMesh — Decentralised Emergency Communication System

Google Nearby Connections API · OpenStreetMap (osmdroid) · MVVM · Background Services

- Engineered P2P mesh networking with automatic device discovery and multi-hop relay via Google Nearby Connections API, eliminating server dependency during outages.
- Built a risk-tiered SOS alert engine with 3 severity levels and auto-propagation across all connected devices, reducing manual coordination in emergencies.
- Integrated OpenStreetMap (osmdroid) for fully offline maps with real-time GPS tracking and radius-based alert zones, enabling location-aware response without internet.
- Validated 100% message delivery across 10 devices within a 25-metre offline mesh under zero-connectivity conditions.

Automated AI Sentry Gun — Computer Vision Targeting System

Python · YOLOv8 · OpenCV · Arduino · PyFirmata2 · Tkinter

- Developed a real-time detection and tracking system using YOLOv8 + OpenCV with low-latency inference, enabling continuous multi-object monitoring at live video frame rates.
- Built a face-recognition training pipeline for personalised individual lock-on, extending capability beyond generic object detection to specific target identification.
- Integrated Arduino pan-tilt servo hardware via PyFirmata2, translating bounding-box coordinates into physical motor commands for smooth real-time tracking.
- Created a Tkinter operator panel consolidating detection modes, training workflows, and hardware controls, reducing mode-switch time to a single click.

ACTIVITIES & CERTIFICATIONS

Runner-Up — TechSangram Hackathon

- Placed 2nd presenting the AI Sentry Gun with a live computer-vision + Arduino integration demo, competing against teams from multiple institutions.

PR Manager, Pichavaram House — IIT Madras

- Managed communications and outreach for student-led house initiatives, coordinating cross-team events and increasing engagement across the cohort.

NPTEL — Introduction to Machine Learning

- Completed supervised/unsupervised learning, model evaluation, and core ML concepts through IIT's nationally recognised online programme.

NPTEL — Data Mining for Decision Making

- Covered data preprocessing, classification, clustering, association rules, and decision-tree methods through IIT's accredited programme.